

HOW TO CHOOSE A SOLID STATE DRIVE FOR YOUR SERVER

Looking to make your server even more efficient? Fit in the right solid state drive (SSD) that can ensure the safety and speedy processing of your data. Here's our guide on how you can buy the best SSD for your server.

As businesses are struggling with the chunks of data to be stored and processed, the traditional storage options are turning obsolete. Gone are the days when solid state drives (SSDs) were restricted to consumer devices. Now, they are increasingly being used with servers, replacing the traditional hard disk drives, and facilitating improved storage performance in tandem with the development of server processors.

IT managers, across the globe, are considering SSDs for servers instead of the traditional storage options. Not only are these more technologically advanced, they also help organisations stay ahead when it comes to data management systems. Although there is a high cost of deployment involved with SSDs, companies are exploring this option of storage over others, particularly in cases of specific workloads. SSDs are especially helpful in cases where speed of data access is critical, like in the e-commerce sector.

Speaking about SSDs for servers, David Chua, director of the distribution business, Asia Pacific South, HGST, says, "SSDs have become the solution of choice to enhance storage I/O throughput and accelerate application performance while reducing latency. These may be deployed at different locations in the data centre — at the server, network or for storage — as either cache or persistent storage to accelerate application performance. In host servers, SSDs with SAS, SATA or PCIe form factors can serve as local cache or storage. Network-based appliances use SSDs as inline cache or Tier 1 storage to speed up disk arrays on the back end. SSDs can also serve as a high-performance tier in a hybrid array, or as a secondary cache behind DRAM in a storage controller."

SSDs in servers are capable of delivering ultra high performance input/output operations. They can provide very low latency for transaction-intensive server and storage applications. Since SSDs involve low operating temperatures and low power consumption, their total cost of ownership is comparatively lower than the traditional storage options. Although SSDs in servers is still a developing phenomenon, IT managers are warming up to the concept and the SSD makers are venturing into the segment in a big way. So, if you are also planning to introduce SSDs to your servers, or even if you have been using them but are not sure if you are making a worthy investment, we will help you out. Here are some key factors that should be kept in mind by IT managers while buying SSDs for servers.

Your investment

Although money is not the first determinant when it comes to buying technology, in the case of SSDs, it surely is an important criteria. These are expensive devices; so spend depending upon the storage capacity you opt for. Obviously, the higher you go on the storage front, the more money you have to shell out. Like a 128GB SSD may cost around Rs 7,000 while a 256GB SSD will fall in the price range of Rs 13,000. So, evaluating prices is an important step.

Storage needs

Before you invest in an SSD, you must remember that you cannot continue with your 'hard-drive centric approach', according to which, the more the storage space, the better it is. Since you have to shell out big money for these devices, you should actually know your storage requirements. For this, you first need to figure out what mission critical applications you want to run. Utilities like SpaceSniffer can be used to determine the storage needs. Once you know these details, you have a fair idea of the storage space that these apps require. It is good to keep some reserve space but it would not be wise to spend on a 512GB SSD if you can fit in your applications in 50GB of space. You must also keep in the mind that the actual space in an SSD is seldom equivalent to what has been advertised. For instance, an SSD that is advertised as having 256GB of space, may just have 240GB in reality. So keep that provision in mind as well, before you make your choice.

SATA II or SATA III?

SATA refers to the serial ATA interface. All the SSDs use this. There are two versions of SATA being used by manufacturers right now - SATA II and SATA III. While SATA II can transfer data at the rate of 3Gbps, SATA III provides a data speed of 6Gbps. Of course, the higher the speed of data transfer, the better it is. SATA III should be your preferred choice, provided the server hardware supports it. SATA II SSDs are cheaper, for obvious reasons. So, you will have to decide whether you want to compromise on speed for the sake of price, or whether burning a hole in your pocket once in order to get a faster product will work for you.

Performance

Let's be sure of one thing first... SSDs are mostly good performers, particularly if you compare them with hard